On (0+1)d SPT interfaces

Tycho Van Camp

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Abstract

We study the $(0+1)$d interfaces of symmetry-protected topological (SPT) chains in the tensor network (TN) framework. By analysing the symmetry fractionalization on the entanglement degrees of freedom, we obtain an algebra with possible anomaly. The representations of this algebra possibly possesses measurable fractional statistics,

Introduction

Boundaries of generalized SPT chains

Outline: - MPO symmetries: first describe how generalized symmetries are presented in the tensor network framework by possible non-local MPOs, the multiplication of MPOs - fusion tensors: fusion tensors are used for the multiplication of MPOs, specifically the multiplication does not have to be group-like. - Action tensors: encode how MPO symmetries act on the entanglement of a state - L symbols: encode the multiplication of symmetry operators on the entanglement of a state, i.e. how does the multiplication of the two action tensors decompose as a sum of action tensors? Most of this will be done in the preliminaries section

Having introduced the tensor network formalism, specifically the representation of symmetry operators, and states, we now proceed to investigate the $(0+1)$d interface of two adjacent SPTs with generalized symmetry encoded in $\mathcal{C}$. Consider two contigous $\mathcal{C}$-SPTs, SPT_A and SPT_B, with corresponding uniform $\mathcal{C}$-symmetric injective MPSs, A and B. Neither the physical- or the virtual Hilbert spaces of A are necessarily isomorphic to that of B. Label the sites of the infinite chain by $i \in \mathbb{Z}$, with $i < 0$ furnished with A tensors, and with $i > 0$ furnished with B tensors. Graphically, we consider the following situation:

[insert MPS chain with A on the left and B on the right, interface is a star]

[Note to self: I have postponed the mention of the action tensors and corresponding L-symbols, since it seems a better 'story'/ 'buildup' of first 'just doing

it' and then reintroducing the previously developed 'tools'.]

Consider the action of an MPO $\mathcal{O}_x$ on this interface:

[insert MPS chain with MPO acting upon it]

Recall that action tensors are used for the local application of an MPO $\mathcal{O}_x$ to an MPS [**?**]. Given the action tensors, $({}^A\phi_x)_i$ and $({}^B\phi_x)_j$, for each of the SPT phases A and B, this simplifies to:

[insert MPS with MPO acting only over boundary]

The operators on the boundary form what is called an *interface algebra* [IO26a]. In the present case, we have a $(0+1)$d interface algebra. Specifically, when SPT_A occupies the left-half-, and SPT_B occupies the right-half of the infinite chain, we have

$$ {}^{A|B}\mathfrak{I} := \{({}^{A|B}\mathcal{I}_x)_{ij} | x \in \mathcal{C}, i, j = 0, 1, ..., \dim(x)\} \tag{1} $$

The identity operator $\mathcal{O}_{\mathbb{I}}$ provides the unit element of ${}^{A|B}\mathfrak{I}$, whereas the muliplication is given by:

[insert multiplication of two interface algebra elements graphically, showing the steps to get to the formula which uses the L symbols]

It was found in [IO26a] that SPT_A and SPT_B belong to the same phase[1] if and only if the interface algebra ${}^{A|B}\mathfrak{I}$ has 1-dimensional representations. More interestingly however, is the case when SPT_A is not in the same phase as SPT_B, as this is the case where degenerate interface modes occur. The existence of these degenerate interface modes[2] is recognized as a hallmark signature of SPTs [FTA25]. This can again be traced back to the fractionalization of the symmetry on the entanglement of the state in question [PBTO12]. These interface modes can obey fractional statistics, and a detection procedure has been outlined in [RVHB19] [GRS25]. This is also found in [SS24] for a $\mathbb{Z}_2$-cluster state and more generally in [FTA25] for the G-cluster state.

Example 1. As a first example, we consider the case where the $\mathbb{Z}_2$-cluster state [SAV11] [Bre15] occupies the left-half of an infinite chain, and the product state occupies the right-half of the infinite chain. Previously, we found that both states have symmetry described by $\mathbf{Vec}_{\mathbb{Z}_2} \boxtimes \mathbf{Rep}(\mathbb{Z}_2) \cong \mathbf{Vec}_{\mathbb{Z}_2} \boxtimes \mathbf{Vec}_{\mathbb{Z}_2}$ [FTA25] [IO26a].
The one-dimensional action tensors of the $\mathbb{Z}_2$-cluster state are: $({}^{\text{cluster}}\phi_g) = V_g$,

[1] Two systems are said to be in the same phase if their corresponding Hamiltonians can be connected via a smooth gapped path of local and symmetric Hamiltonians [GRLM23].

[2] We distinguish between edge modes when the protecting symmetry is group-based, versus interface modes when the protecting symmetry is a generalized symmetry.

with V_g given in table 1. The L-symbols evaluate to a normalised representative ω in the non-trivial second cohomology class of $\mathbb{Z}_2 \times \mathbb{Z}_2$ in $U(1)$), denoted $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U(1))$: ${}^{\text{cluster}}L_{gh} = \omega(g,h)$. The action tensors of the product state are trivial. Similarly, the L-symbols of the product state are trivial.

g	V_g
(0,0)	$\mathbb{I}_2\mathbb{I}_2$
(0,1)	$\mathbb{I}_2 Z$
(1,0)	$X\mathbb{I}_2$
(1,1)	XZ

Table 1: Projective representations for the $\mathbb{Z}_2$-cluster state. $\mathbb{I}_2$ is the 2×2 identity matrix, and X, Y, Z are the usual Pauli matrices.

Using the fact that the $\mathbb{Z}_2$-cluster state and the trivial product state are not in the same phase (which should be clear from the inequivalence[3] of L-symbols), and that the interface algebra ${}^{\text{cluster}|\text{triv}}\mathfrak{I}$ has total dimension $|\mathbb{Z}_2 \times \mathbb{Z}_2| = 4$, we conclude that ${}^{\text{cluster}|\text{triv}}\mathfrak{I} \cong \text{Mat}_2(\mathbb{C})$. The algebra of 2×2 matrices over $\mathbb{C}$, $\text{Mat}_2(\mathbb{C})$, is spanned by the usual Pauli matrices: $\{\mathbb{I}_2, \sigma^x, \sigma^y, \sigma^z\}$ [note to self: notice the different notation used for the interface algebra and the action tensors, or the bulk symmetry operators]. Multiplication of operators in the interface algebra ${}^{\text{cluster}|\text{triv}}\mathfrak{I}$ reads:

$$ {}^{\text{cluster}|\text{triv}}\mathcal{I}_g \times {}^{\text{cluster}|\text{triv}}\mathcal{I}_h = \omega(g,h)\,{}^{\text{cluster}|\text{triv}}\mathcal{I}_{gh} \tag{2} $$

Specific operators ${}^{\text{cluster}\ |\text{triv}}\mathcal{I}_g$ in the interface algebra ${}^{\text{cluster}|\text{triv}}\mathfrak{I}$ for $g \in \mathbb{Z}_2 \times \mathbb{Z}_2$ are given in table 2.

g	${}^{\text{cluster}\ |\text{triv}}\mathcal{I}_g$
$(0,0)$	$\mathbb{I}_2$
$(0,1)$	σ^z
$(1,0)$	σ^x
$(1,1)$	$i\sigma^y$

Table 2: Representation of interface algebra operators ${}^{\text{cluster}\ |\text{triv}}\mathcal{I}_g$ in ${}^{\text{cluster}|\text{triv}}\mathfrak{I}$ for $g \in \mathbb{Z}_2 \times \mathbb{Z}_2$. The $\sigma^x, \sigma^y, \sigma^z$ are the usual 2×2 Pauli matrices, and $\mathbb{I}_2$ is the 2×2 identity matrix.

Next, we provide an example of a $(0+1)$d interface algebra between SPT phases protected by a non-invertible symmetry. In contrast to the case of invertible SPT phases where the identification of a trivial product state is clear-cut, the notion of a trivial product state for non-invertible SPT phase does not exist [SS24].

[3] We say that two sets of L-symbols are equivalent if they are related by gauge transformations on the action tensors whence they are defined [GRLM23] [IO26a].

Example 2. In this example we consider the interface algebras between different $\mathbf{Rep}(D_4)$ SPT phase where D_4 is the group describing the symmetries of a square. We provide the fusion of irreducible representations of D_4 in table 3.

$\otimes$	A_1	A_2	B_1	B_2	E
A_1	A_1	A_2	B_1	B_2	E
A_2		A_1	B_2	B_1	E
B_1			A_1	A_2	E
B_2				A_1	E
E					$A_1 \oplus A_2 \oplus B_1 \oplus B_2$

Table 3: Fusion rules of irreps of the dihedral group D_4. The empty slots can be found by transposing the table. Notice that all of the fusion channels yield multiplicity-free products.

It is noteworthy to mention that $\mathbf{Rep}(D_4)$ is a group-theoretical fusion 1-category [Nat17], such that explicit lattice models of these phases can be constructed [SS24]. Specifically, these models are obtained by starting from a D_4-symmetric Hamiltonian, for which the ground state (GS) has spontaneous symmetry breaking (SSB) pattern $G \to K$, and subsequently applying (twisted) gauging [HVAS$^+$15] [GRK23] [CGRV$^+$25] of the subgroup $H \subseteq G$, such that K and H provide an exact factorization of G [SS24][IO26b]. Alternatively, starting from the different $\mathbf{Rep}(D_4)$-SPT models and subsequently applying the (twisted) gauging in reverse, one observes different SSB patterns of the resulting SSB models. It is know that $\mathbf{Rep}(D_4)$ admits 3 distinct non-invertible SPT phases[4] [Tam00]. We will label the distinct $\mathbf{Rep}(D_4)$-SPT phases by $\mathrm{SPT}_{\mathsf{A}}$, $\mathrm{SPT}_{\mathsf{B}}$, and $\mathrm{SPT}_{\mathsf{C}}$. [Note to self: it might be interesting to write down all three SPT models in TN framework following [SS24], and subsequently extracting the action tensors. For now, we'll trust [IO26a].] The L-symbols for the $\mathbf{Rep}(G)$-SPT phases boil down to (twisted) generalized Clebsch-Gordan coefficients[5] (GCGCs) [IO26a]. We list the nontrivial ${}^{\mathsf{A}}L$-symbols for $\mathrm{SPT}_{\mathsf{A}}$ in table 4.

For the other two SPT phases, $\mathrm{SPT}_{\mathsf{B}}$ and $\mathrm{SPT}_{\mathsf{C}}$, we have a *twisted* copy of the ${}^{\mathsf{A}}L$-symbols of $\mathrm{SPT}_{\mathsf{A}}$, wherein another one of the L-symbols regarding the fusion of a one-dimensional irrep (A_i or B_i) and the two-dimensional irrep (E), changes sign when interchanging the reactants. The ${}^{\mathsf{B}}L$-symbols for $\mathrm{SPT}_{\mathsf{B}}$ are provided in table 5, whilst the ${}^{\mathsf{C}}L$-symbols for $\mathrm{SPT}_{\mathsf{C}}$ are provided in table 6.

With the L-symbols at hand, we can now investigate the interface algebras between these non-invertible SPTs. Let L denote the SPT phase occupying the left-half of the infinite chain, and R denote the right-half of the infinite chain.

[4] Distinct SPT phases with symmetry encoded in the fusion 1-category $\mathcal{C}$ are provided by non-isomorphic fiber 1-functors [TW19].

[5] Generalized Clebsch-Gordan coefficients of small groups can be found in [KDWS63], and a formula for computing these for groups enjoying multiplicity-free fusion of unitary irreducible representation is given in [Aro26]. A numerical algorithm applicable to compact groups can be found in [ILYM17].

$({}^{\mathsf{A}}L^{E}_{EA_1})^{k}_{1j} = ({}^{\mathsf{A}}L^{E}_{A_1E})^{k}_{1j} = \left[\begin{smallmatrix} A_1 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\mathbb{I}_2)_{jk}$	$({}^{\mathsf{A}}L^{A_1}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} A_1 \\ 1 \end{smallmatrix}\right] = (\mathbb{I}_2)_{ij}$
$-({}^{\mathsf{A}}L^{E}_{EA_2})^{k}_{1j} = ({}^{\mathsf{A}}L^{E}_{A_2E})^{k}_{1j} = \left[\begin{smallmatrix} A_2 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (-\mathsf{i}\sigma^y)_{jk}$	$({}^{\mathsf{A}}L^{A_2}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} A_2 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(\mathsf{i}\sigma^y)_{ij}$
$({}^{\mathsf{A}}L^{E}_{EB_1})^{k}_{1j} = ({}^{\mathsf{A}}L^{E}_{B_1E})^{k}_{1j} = \left[\begin{smallmatrix} B_1 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\sigma^z)_{jk}$	$({}^{\mathsf{A}}L^{B_1}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} B_1 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(\sigma^z)_{ij}$
$({}^{\mathsf{A}}L^{E}_{EB_2})^{k}_{1j} = ({}^{\mathsf{A}}L^{E}_{B_2E})^{k}_{1j} = \left[\begin{smallmatrix} B_2 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\sigma^x)_{jk}$	$({}^{\mathsf{A}}L^{B_2}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} B_2 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(\sigma^x)_{ij}$

Table 4: Nontrivial ${}^{\mathsf{A}}L$-symbols for the $\mathbf{Rep}(D_4)$-SPT phase $\mathrm{SPT}_{\mathsf{A}}$ [KDWS63] [IO26a]. $\sigma^x, \sigma^y, \sigma^z$ are the usual 2×2 Pauli matrices. Note that $({}^{\mathsf{B}}L^{E}_{A_2E})^{k}_{1j} = -({}^{\mathsf{B}}L^{E}_{EA_2})^{k}_{1j}$. [Note to self: We identify $\mathrm{SPT}_{\mathsf{A}}$ with the ν_3 SPT phase in [IO26a]]

$({}^{\mathsf{B}}L^{E}_{EA_1})^{k}_{1j} = ({}^{\mathsf{B}}L^{E}_{A_1E})^{k}_{1j} = \left[\begin{smallmatrix} A_1 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\mathbb{I}_2)_{jk}$	$({}^{\mathsf{B}}L^{A_1}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} A_1 \\ 1 \end{smallmatrix}\right] = (\sigma^x)_{ij}$
$({}^{\mathsf{B}}L^{E}_{EA_2})^{k}_{1j} = ({}^{\mathsf{B}}L^{E}_{A_2E})^{k}_{1j} = \left[\begin{smallmatrix} A_2 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (-\mathsf{i}\sigma^y)_{jk}$	$({}^{\mathsf{B}}L^{A_2}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} A_2 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(\sigma^z)_{ij}$
$-({}^{\mathsf{B}}L^{E}_{EB_1})^{k}_{1j} = ({}^{\mathsf{B}}L^{E}_{B_1E})^{k}_{1j} = \left[\begin{smallmatrix} B_1 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\sigma^z)_{jk}$	$({}^{\mathsf{B}}L^{B_1}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} B_1 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(\mathsf{i}\sigma^y)_{ij}$
$({}^{\mathsf{B}}L^{E}_{EB_2})^{k}_{1j} = ({}^{\mathsf{B}}L^{E}_{B_2E})^{k}_{1j} = \left[\begin{smallmatrix} B_2 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\sigma^x)_{jk}$	$({}^{\mathsf{B}}L^{B_2}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} B_2 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(\mathbb{I}_2)_{ij}$

Table 5: Nontrivial ${}^{\mathsf{B}}L$-symbols for the $\mathbf{Rep}(D_4)$-SPT phase $\mathrm{SPT}_{\mathsf{B}}$ [IO26a]. $\sigma^x, \sigma^y, \sigma^z$ are the usual 2×2 Pauli matrices. Note that $({}^{\mathsf{B}}L^{E}_{B_1E})^{k}_{1j} = -({}^{\mathsf{B}}L^{E}_{EB_1})^{k}_{1j}$. [Note to self: We identify $\mathrm{SPT}_{\mathsf{B}}$ with the ν_1 SPT phase in [IO26a]]

The eight-dimensional interface algebra ${}^{\mathsf{L}j\mathsf{R}}\mathfrak{I}$ is spanned by:

$$\{{}^{\mathsf{L}|\mathsf{R}}\mathcal{I}_{A_i}, {}^{\mathsf{L}|\mathsf{R}}\mathcal{I}_{B_j}, ({}^{\mathsf{L}|\mathsf{R}}\mathcal{I}_{E})_{ij} | i, j = 1, 2\}. \tag{3}$$

When L is not in the same SPT phase as R, we again use the result from [IO26a] which determines that there exist no one-dimensional representations of the ${}^{\mathsf{L}|\mathsf{R}}\mathfrak{I}$, to conclude that it must be the case that ${}^{\mathsf{L}|\mathsf{R}}\mathfrak{I} \cong \mathrm{Mat}_2(\mathbb{C}) \oplus \mathrm{Mat}_2(\mathbb{C})$. This also implies a four-fold degeneracy at the interface between different $\mathbf{Rep}(D_4)$-SPTs. Multiplication of operators in ${}^{\mathsf{L}j\mathsf{R}}\mathfrak{I}$ is given by

Summary

$({}^{\mathsf{C}}L^{E}_{EA_1})^{k}_{1j} = ({}^{\mathsf{C}}L^{E}_{A_1E})^{k}_{1j} = \left[\begin{smallmatrix} A_1 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\mathbb{I}_2)_{jk}$	$({}^{\mathsf{C}}L^{A_1}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} A_1 \\ 1 \end{smallmatrix}\right] = (\sigma^z)_{ij}$
$({}^{\mathsf{C}}L^{E}_{EA_2})^{k}_{1j} = ({}^{\mathsf{C}}L^{E}_{A_2E})^{k}_{1j} = \left[\begin{smallmatrix} A_2 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (-\mathsf{i}\sigma^y)_{jk}$	$({}^{\mathsf{C}}L^{A_2}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} A_2 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(-\sigma^x)_{ij}$
$({}^{\mathsf{C}}L^{E}_{EB_1})^{k}_{1j} = ({}^{\mathsf{C}}L^{E}_{B_1E})^{k}_{1j} = \left[\begin{smallmatrix} B_1 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\sigma^z)_{jk}$	$({}^{\mathsf{C}}L^{B_1}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} B_1 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(\mathbb{I}_2)_{ij}$
$-({}^{\mathsf{C}}L^{E}_{EB_2})^{k}_{1j} = ({}^{\mathsf{C}}L^{E}_{B_2E})^{k}_{1j} = \left[\begin{smallmatrix} B_2 & E \\ 1 & j \end{smallmatrix}\middle\vert\begin{smallmatrix} E \\ k \end{smallmatrix}\right] = (\sigma^x)_{jk}$	$({}^{\mathsf{C}}L^{B_2}_{EE})^{1}_{ij} = \left[\begin{smallmatrix} E & E \\ i & j \end{smallmatrix}\middle\vert\begin{smallmatrix} B_2 \\ 1 \end{smallmatrix}\right] = \frac{1}{\sqrt{2}}(-\mathsf{i}\sigma^y)_{ij}$

Table 6: Nontrivial ${}^{\mathsf{C}}L$-symbols for the $\mathbf{Rep}(D_4)$-SPT phase $\mathrm{SPT}_{\mathsf{C}}$ [IO26a]. $\sigma^x, \sigma^y, \sigma^z$ are the usual 2×2 Pauli matrices. Note that $({}^{\mathsf{C}}L^{E}_{B_2E})^{k}_{1j} = -({}^{\mathsf{C}}L^{E}_{EB_2})^{k}_{1j}$. [Note to self: We identify $\mathrm{SPT}_{\mathsf{C}}$ with the ν_2 SPT phase in [IO26a]]

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